

Calculus 3 Problems

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Problem 1

Consider the region

$$\Omega := \{(x, y, z) : x^2 + y^2 + z^2 \leq 4\}$$

Show that the volume of the region is $\frac{256\pi}{3}$ using triple integration.

Problem 2

Suppose the magnetic field generated by an electromagnet is given by

$$\mathbf{B}(x, y) = \frac{\mu_0 I}{2\pi(x^2 + y^2)} \langle -y, x, 0 \rangle$$

Find the net work done by this field along the path

$$\mathbf{r}(t) = \langle t^3, \cos(t), \sin(t) \rangle$$

from the point $(0, 1, 0)$ to $(\pi^3/8, 0, 1)$.

Suppose the particle now takes on the path:

$$\mathbf{v}(t) = \langle \cos(2\pi t), 2\sin(2\pi t), 0 \rangle$$

Show that the net work done along this path by the magnetic field is zero. Explain, physically, why this is the case.

Problem 3

A robot must travel from point $(0, 0)$ to point $(1, 1)$ while staying on a smooth curve C . The energy this robot requires to traverse the path is given by:

$$E[C] = \int_C x^2 + y^2 ds$$

where ds is the arc length differential element.

However, the robot can only follow paths, whose curvature $\kappa(s)$ satisfies the constraint $\kappa(s) \leq 1$ at every point along the curve. Determine the minimum possible energy required by the robot.

Problem 4

A UAV manufacturing company is calibrating a sensor, whose true response is given by the expression $s(t) = e^{-\frac{1}{2}t^2}$ on the interval $t \in [0, 1]$. However, due to hardware limitations such as heating of components, computing power, and power supply limitations, the sensor can only implement a linear approximation of the data, $\hat{s}(t) = \alpha t + \beta$. The company defines the loss function to be the total squared error accumulated over the entire operating interval. Determine the calibration parameters that produce the most accurate sensor.

Hint: you may encounter some difficult integrals, for which you may leave the answer as values of the error function:

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$$

So in any case where you encounter integrals such as $\int_0^1 e^{-t^2} dt$, you may simply leave it as $\frac{\sqrt{\pi}}{2} \text{erf}(1)$.

Problem 5

Use Taylor expansions to approximate the value of the following integral to 3 decimal places:

$$\int_0^2 e^{-\frac{1}{2}t^3} dt$$